2026 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)
13-14 August 2026, Faculty of ECE, Bangladesh Army University of Engineering & Technology (BAUET), Qadirabad, Natore-6431, Bangladesh

Secure Outage Performance Analysis of Reconfigurable Intelligent Surface Assisted Communication System Over κ-μ Shadowed Fading Channels

M. A. Chowdhury, S. K. Safit, M. Z. I. Sarkar, Ahshan Habib, Md Najmul Huda
Department of Electrical & Electronic Engineering, RUET, Rajshahi-6204, Bangladesh
Email: abidchy303@gmail.com; shahziarsafit074@gmail.com; msarkar@eee.ruet.ac.bd;
ahshanhabib581@gmail.com; mdnajmulhuda555555@gmail.com

***Abstract*—This paper proposes an RIS-assisted communication framework to evaluate physical layer security (PLS) performance under κ-μ shadowed fading conditions in the presence of an eavesdropper. Since the link between the base station and the destination is obstructed, all communication goes through the RIS, which has N passive reflecting elements. The signals received by the destination and the eavesdropper are modeled using the κ-μ shadowed fading distribution, capturing the combined effects of multipath clustering, dominant line-of-sight components, and shadowing. Series-form analytical expressions for the probability of non-zero secrecy capacity (PNZSC), intercept probability (IP), and secure outage probability (SOP) are formulated and validated using Monte Carlo simulation techniques. The effects of the number of RIS reflecting elements N, Rician factor κ, and multipath clustering parameter μ on the secrecy performance are thoroughly investigated. The results confirm that increasing N, κ and μ significantly enhances the system's security, establishing RIS as an effective approach for securing future wireless networks without increasing transmit power.**

***Index Terms*—Reconfigurable intelligent surface (RIS), physical layer security (PLS), κ-μ shadowed fading, probability of non-zero secrecy capacity (PNZSC), intercept probability (IP), secure outage probability (SOP).**

I. Introduction

Wireless communication systems are susceptible to eavesdropping since radio propagation is inherently broadcast in nature. Conventional security mechanisms operating at upper layers, such as cryptographic protocols, require substantial computational overhead and are becoming more difficult to implement due to the increasingly complex modern networks. Physical layer security (PLS), a fundamentally distinct approach that leverages the randomness of the wireless channels to realize information-theoretic security without computational assumptions [1].

A. Related Works

Recently, reconfigurable intelligent surface (RIS) technology has drawn significant research interest as a resource-efficient and cost-effective approach for enhancing the reliability and security of wireless systems [2]. The RIS comprises a large array of reflecting elements that are capable of controlling the phase of the incident electromagnetic waves independently. Using smart control of the phases of the impinging electromagnetic waves, an RIS system can modify its propagation environment to improve wireless system reliability and security. [3].

Several works considered the performance of PLS scheme in RIS- enabled systems using various channel models. Yang *et al.* [1] derived closed-form formulas for SOP in RIS-enabled networks operating over Rayleigh fading channels. Do *et al.* [4] analyzed secure performance of RIS-aided systems and demonstrated significant secrecy gains over conventional systems. Yadav *et al.* [5] considered the security performance of RIS-enabled networks under Rayleigh fading conditions to obtain PNZSC and SOP for single-input single-output architecture. However, these studies only consider Rayleigh fading channel which fails to represent multipath clustering, dominant paths and signal attenuation due to shadowing.

The κ-μ shadowed fading model [6] offers a realistic approach to wireless environment propagation. It works as a generalization, which encompasses various models of fading, including Rayleigh, Nakagami-m, Rician, and κ-μ. For this reason, it is an effective tool to characterize different wireless propagation scenarios. Recently, Al-Rimawi and Al-Dweik [7] proposed a theoretical model of secure communication scheme with the assistance of RIS and direct link in κ-μ shadowed fading channels using outage probability and BER analysis. The RIS-supported communication network that utilizes energy harvesting (EH) in Nakagami-m fading channel was highlighted in [8]. The authors [9] further investigated the secure characteristics of the RIS-assisted NOMA system under the κ-μ shadowed fading environment along with the outage and interception probabilities. Sikri *et al.* [10] examined the effects of phase error and hardware imperfections on RIS-aided systems under κ-μ fading environment. Mamun *et al.* [11] performed an analysis on the ergodic secrecy capacity of RIS-aided networks for a single eavesdropper scenario. The comparison among the proposed study and chosen previous works is provided in Table I.

979-8-3195-1270-3/26/$31.00 ©2026 IEEE

TABLE I: Comparison with Existing Works

Ref.	Channel	RIS	Security	Metrics
[1]	Rayleigh	✓	✓	SOP
[4]	Rayleigh	✓	✓	SOP
[5]	Rayleigh	✓	✓	SOP, IP, PNZSC
[11]	Rayleigh	✓	✓	ESC
[9]	κ-μ Shadowed	✓	✓	OP, IP
[7]	κ-μ Shadowed	✓	-	OP, BER
[10]	κ-μ	✓	-	CP, EC
This work	**κ-μ Shadowed**	✓	✓	**PNZSC, IP, SOP**

B. Contributions

Even with the above contributions, there still exist some open research questions concerning the PLS analysis of RIS-enabled networks with κ-μ shadowed fading considering PNZSC, IP, and SOP. Particularly, no study has been conducted on the situation where the LOS link is entirely obstructed and all propagation occurs through RIS only under κ-μ shadowed fading conditions. Motivated by this research gap, this study presents the following contributions. The key novelty lies in deriving PNZSC, IP, and SOP for the fully RIS-dependent scenario where the direct BS-to-destination link is completely blocked. This scenario has not been addressed in prior secrecy studies under κ-μ shadowed fading:

- Series-form analytical expressions for PNZSC, IP and SOP are derived from the statistical characterization of both the desired and undesired channels under SISO communication system over κ-μ shadowed fading channels.
- The impacts of the number of reflecting elements N, Rician factor κ, and multipath clustering parameter μ on all three secrecy metrics thoroughly investigated.
- The proposed analytical models are numerically verified through Monte Carlo simulation, demonstrating close agreement between analytical and simulated results.

The structure of the rest of the paper is as follows. The system framework and channel assumptions are introduced in Section II. Section III develops the analytical expressions for the secrecy metrics. IV provides numerical evaluations along with simulation validation. Section V concludes the study.

II. System Model

An RIS-supported wireless communication network is considered, consisting of a source (S), an authorized destination user (D), a passive eavesdropper (E), and an RIS that comprises N passive reflectors deployed on a building, as shown in Fig. 1. Each node works with a single antenna, forming a SISO system. The communication link between the BS and the destination is known as the main channel while the link between the BS and the eavesdropper is known as the eavesdropper channel. The BS sends confidential data to the RIS, which smartly steers the signal to the authorized destination user, despite the presence of the eavesdropper. In the meantime, the eavesdropper tries to intercept and decode the signal sent. The proposed model will improve the PLS by optimizing the quality of the legitimate channel and reducing the information leakage to the eavesdropper and hence secure

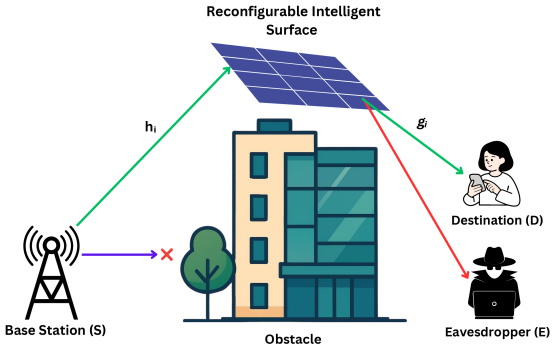


Fig. 1: RIS-assisted communication system between the base station and the destination in the presence of an eavesdropper.

communication between the BS and the destination user. Let h_i and g_i represent the channel coefficients corresponding to the source-to-RIS and RIS-to-destination links for the i-th RIS element, $i = 1, 2, \ldots, N$, and let e_i represent the RIS-to-eavesdropper channel coefficient. The RIS applies phase shift $\phi_i = -(\angle h_i + \angle g_i)$ to ensure coherent combining at D [7]. The signals received at D and E are expressed as

$$y_D = \sqrt{P}\sum_{i=1}^{N} g_i e^{j\phi_i} h_i\, s + n_D, \tag{1}$$

$$y_E = \sqrt{P}\sum_{i=1}^{N} e_i e^{j\phi_i} h_i\, s + n_E, \tag{2}$$

where P is the transmit power, s denotes the transmitted symbol with $\mathbb{E}[|s|^2] = 1$, and n_D, $n_E \sim \mathcal{CN}(0, \sigma^2)$ are the AWGN terms at the destination (D) and the eavesdropper (E) respectively. The instantaneous received SNR at D and E are respectively expressed as

$$\gamma_d = \frac{P}{\sigma^2}\left|\sum_{i=1}^{N} g_i e^{j\phi_i} h_i\right|^2, \tag{3}$$

$$\gamma_e = \frac{P}{\sigma^2}\left|\sum_{i=1}^{N} e_i e^{j\phi_i} h_i\right|^2. \tag{4}$$

The channel links corresponding to source-to-RIS, RIS-to-destination, and RIS-to-eavesdropper are assumed to be independent and follow κ-μ shadowed fading characteristics [6], [7]. This model provides a flexible characterization of wireless channels by explicitly accounting for multipath clusters through the variable μ, the power ratio between dominant rays and scattered waves through κ, and the severity of shadowing on the dominant components through ω. This model subsumes several widely used distributions, including Rayleigh, Nakagami-m, Rician, and κ-μ, as special cases [6].

Based on the statistical framework established in [7], the end-to-end SNR distributions at D and E are characterized by their respective probability density functions (PDFs), which are derived in Section III.

III. Problem Formulation

A. PDF of Received SNR at the Destination User

Let γ_d represent the instantaneous SNR at the destination user. The corresponding distribution under κ-μ shadowed fading is expressed in [7]

$$f_{\gamma_d}(\gamma_d) = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}}{\Gamma(q)\,\alpha^a}\left(\gamma_d^{q-1}\, e^{-\gamma_d/\alpha}\right) {}_1F_1(q_1,\, q;\, \beta\gamma_d), \tag{5}$$

where α, β, a, q, and q_1 denote the scaling parameter, shaping parameter, shape-related normalization constant, effective multipath parameter, and shadowing adjustment parameter, respectively. $\Gamma(\cdot)$ and ${}_1F_1(\cdot)$ denote the Gamma and confluent hypergeometric functions, respectively. The coefficient $G_{i,j}$ is expressed as

$$G_{i,j} = \frac{\sqrt{\pi}\, H_q}{\Gamma(n_q)\,\Omega_q^{n_q-0.5}} \cdot \frac{\lambda_2!\, a_u}{(u-1)!\,\Delta^u},$$

where $\lambda_1 = 2n_q + u - 1$, $\lambda_2 = \lambda_1 - 1$, $a_u \in \{0.25482959, -0.284496736, 1.421413741, -1.453152027, 1.061405429\}$ for $u = 1, \ldots, 5$, $\Delta = \frac{r}{2}\sqrt{\Omega_q}$, $r = 0.3275911$, and

$$\Omega_q = \frac{\mu\kappa + \omega}{\omega} \cdot \frac{1}{\mu(1+\kappa)}.$$

Here, M denotes the number of terms used in the summation-based representation of the κ-μ shadowed PDF, and the coefficients a_u together with $r = 0.3275911$ arise from the Abramowitz-Stegun rational approximation of the complementary error function used to evaluate $G_{i,j}$ [12]. Using the series expansion of ${}_1F_1(\cdot)$ via [13, eq. 9.14(1)], the confluent hypergeometric term is simplified as

$${}_1F_1(q_1,\, q;\, \beta\gamma_d) = \sum_{k_1=0}^{\infty} B_4(k_1)\,(\gamma_d)^{k_1}, \tag{6}$$

where

$$B_4(k_1) = \frac{\Gamma(q_1+k_1)\,\Gamma(q)\,\beta^{k_1}}{\Gamma(q_1)\,\Gamma(q+k_1)\,k_1!}.$$

Substituting (6) into (5) gives the simplified PDF as

$$f_{\gamma_d}(\gamma_d) = B_1 \sum_{k_1=0}^{\infty} B_4(k_1)\,\gamma_d^{(B_2+k_1)}\, e^{-B_3\gamma_d}, \tag{7}$$

where

$$B_1 = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}}{\Gamma(q)\,\alpha^a}, \quad B_2 = q-1, \quad B_3 = \frac{1}{\alpha}.$$

B. PDF of Received SNR at the Eavesdropper

Let γ_e denotes the instantaneous SNR at the eavesdropper. The PDF $f_{\gamma_e}(\gamma_e)$ for the κ-μ shadowed fading channel follows the same structure as (5) [7], expressed as

$$f_{\gamma_e}(\gamma_e) = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}}{\Gamma(q)\,\alpha^a}\left(\gamma_e^{q-1}\, e^{-\gamma_e/\alpha}\right) {}_1F_1(q_1,\, q;\, \beta\gamma_e). \tag{8}$$

Using [13, eq. 9.14(1)], the hypergeometric term simplifies as

$${}_1F_1(q_1,\, q;\, \beta\gamma_e) = \sum_{k_2=0}^{\infty} C_4(k_2)\,(\gamma_e)^{k_2}, \tag{9}$$

where

$$C_4(k_2) = \frac{\Gamma(q_1+k_2)\,\Gamma(q)\,\beta^{k_2}}{\Gamma(q_1)\,\Gamma(q+k_2)\,k_2!}.$$

Substituting (9) into (8) gives

$$f_{\gamma_e}(\gamma_e) = C_1 \sum_{k_2=0}^{\infty} C_4(k_2)\,\gamma_e^{(C_2+k_2)}\, e^{-C_3\gamma_e}, \tag{10}$$

where

$$C_1 = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}}{\Gamma(q)\,\alpha^a}, \quad C_2 = q-1, \quad C_3 = \frac{1}{\alpha}.$$

C. Probability of Non-Zero Secrecy Capacity

The PNZSC is defined as [14]

$$\Pr(C_s > 0) = \int_0^{\infty} f_{\gamma_d}(\gamma_d) \left[\int_0^{\gamma_d} f_{\gamma_e}(\gamma_e)\, d\gamma_e\right] d\gamma_d. \tag{11}$$

Substituting (7) and (10) into (11) and applying [13, eq. 3.351(1), eq. 3.351(3)], the series-form analytical expression is obtained as

$$\begin{aligned}\Pr(C_s > 0) = B_1 C_1 \sum_{k_1=0}^{\infty}\sum_{k_2=0}^{\infty} B_4(k_1)\, C_4(k_2) \\ \times \left[\frac{(C_2+k_2)!\,(B_2+k_1)!}{C_3^{C_2+k_2+1}\, B_3^{(B_2+k_1+1)}} \right. \\ \left. - D_1(k_1,k_2)\right],\end{aligned} \tag{12}$$

where

$$D_1(k_1,k_2) = \sum_{k_5=0}^{C_2+k_2} \frac{(C_2+k_2)!\,(B_2+k_1+k_5)!}{k_5!\, C_3^{C_2+k_2-k_5+1}\,(B_3+C_3)^{(B_2+k_1+k_5+1)}}.$$

D. Intercept Probability

The IP is defined as [5]

$$\Pr(C_s < 0) = 1 - \int_0^{\infty} f_{\gamma_d}(\gamma_d) \left[\int_0^{\gamma_d} f_{\gamma_e}(\gamma_e)\, d\gamma_e\right] d\gamma_d. \tag{13}$$

Substituting (12) into (13), the series-form IP is

$$\begin{aligned}\Pr(C_s < 0) = 1 - B_1 C_1 \sum_{k_1=0}^{\infty}\sum_{k_2=0}^{\infty} B_4(k_1)\, C_4(k_2) \\ \times \left[\frac{(C_2+k_2)!\,(B_2+k_1)!}{C_3^{C_2+k_2+1}\, B_3^{(B_2+k_1+1)}} \right. \\ \left. - D_1(k_1,k_2)\right],\end{aligned} \tag{14}$$

E. Secure Outage Probability

The SOP is defined as [14]

$$P_{\text{out}}(R_s) = 1 - \int_0^{\infty} f_{\gamma_e}(\gamma_e) \left[\int_x^{\infty} f_{\gamma_d}(\gamma_d)\, d\gamma_d \right] d\gamma_e, \quad (15)$$

where $x = e^{2R_s}(1+\gamma_e) - 1$. Applying [13, eq. 3.351(2)], the series-form SOP is

$$P_{\text{out}}(R_s) = 1 - B_1 C_1\, B_6 \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} \sum_{k_6=0}^{B_2+k_1} \sum_{k_7=0}^{k_6} B_4(k_1)\, C_4(k_2) \times B_5(k_6, k_7)\, (D_3!)\, (C_3 + B_7)^{-(D_3+1)}, \quad (16)$$

where

$$B_5(k_6, k_7) = \binom{k_6}{k_7} \left(e^{2R_s} - 1\right)^{k_6 - k_7} e^{2R_s k_7} \frac{(B_2 + k_1)!}{k_6!\, B_3^{B_2+k_1-k_6+1}},$$

$$B_6 = e^{-B_3(e^{2R_s}-1)},$$

$$B_7 = B_3\, e^{2R_s},$$

$$D_3 = C_2 + k_2 + k_7.$$

IV. Numerical Results

In this section, the analytical results for PNZSC, IP, and SOP are computed using Mathematica from the derived formulations. The infinite series in $B_4(k_1)$ and $C_4(k_2)$ are truncated to $K = 20$ terms, which was verified to be sufficient for convergence within an accuracy of 10^{-6} across all considered parameter ranges. Simulations for this model are conducted in MATLAB using 100,000 Monte Carlo iterations to generate the numerical results. All the figures have been plotted using the obtained expressions. The excellent match observed in all the figures confirms the validity of the obtained expressions. Unless otherwise stated, $N = 2$ is used as a baseline value in Figs. 3 and 4 to isolate the influence of κ and μ without the result being dominated by the RIS beamforming gain, whereas $N = 4, 6, 8$ is used in the remaining figures to specifically highlight the influence of the number of RIS elements.

Fig. 2 illustrates the impact of the RIS elements count N on PNZSC versus main channel SNR. It is observed that raising N from 4 to 8 substantially enhances PNZSC throughout the entire SNR range. This is because a larger number of reflecting elements provides stronger beamforming gain toward D, which increases γ_d while the eavesdropper SNR γ_e remains comparatively weaker.

Fig. 3 presents the impact of the Rician parameter κ on PNZSC. With an increase in the value of κ from 6 to 10, there is a considerable enhancement in PNZSC. An increase in κ implies that the strength of the dominant line-of-sight (LOS) component becomes more significant relative to the scattered components, resulting in improved SNR at point D and increasing the chance of obtaining a positive secrecy rate. Consequently, the main channel becomes more robust against fading, resulting in better-quality signals received by the desired user.

Fig. 4 shows the influence of the multipath cluster factor μ on PNZSC. Increasing μ from 4 to 6 leads to a consistent improvement in PNZSC. A larger μ corresponds to a greater number of multipath clusters arriving at the receiver, which contributes to higher diversity and a more stable received signal at D.

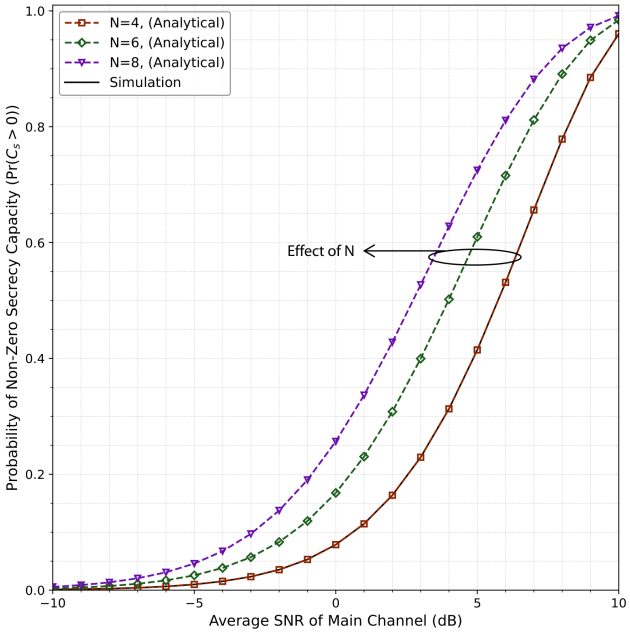


Fig. 2: The effect of N on PNZSC for $\kappa = 1$, $\mu = 1$ and $\bar{\gamma}_e = 1$dB.

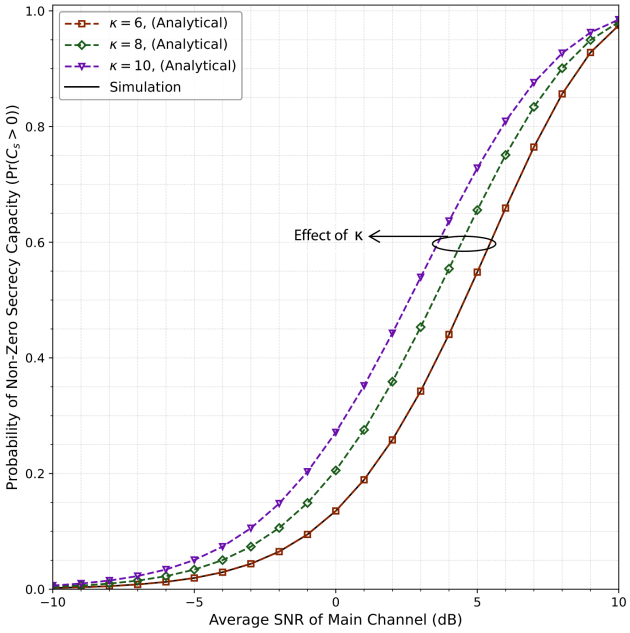


Fig. 3: The effect of κ on PNZSC when N=2, and $\bar{\gamma}_e$=1 dB.

Fig. 5 depicts the IP versus the main channel SNR for varying N. As expected, the IP decreases with increasing SNR, indicating improved secrecy at higher power levels. Additionally, increasing N from 4 to 8 results in a considerable drop in IP, confirming that a larger RIS effectively suppresses

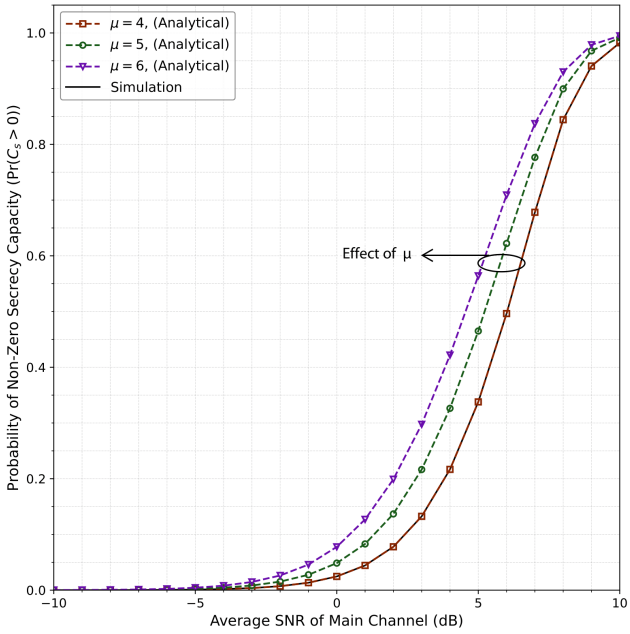


Fig. 4: The impact of μ on PNZSC when N=2, and $\bar{\gamma}_e$=1 dB.

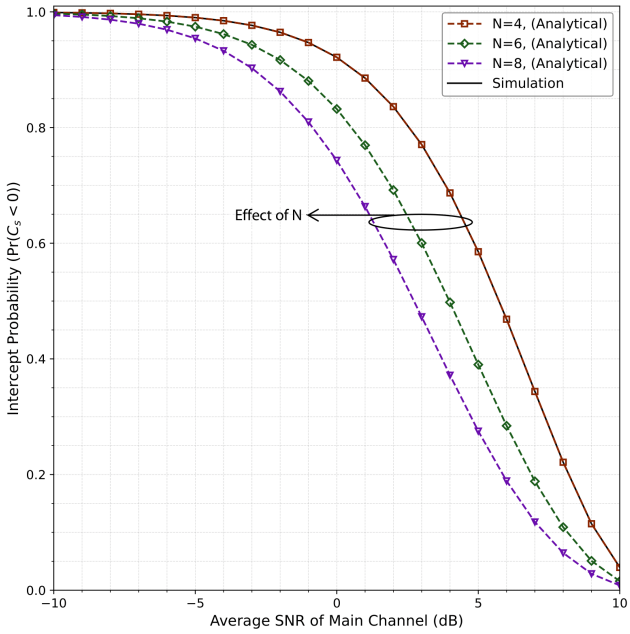


Fig. 5: The effect of N on IP for $\kappa = 1$, $\mu = 1$ and $\bar{\gamma}_e = 1$dB.

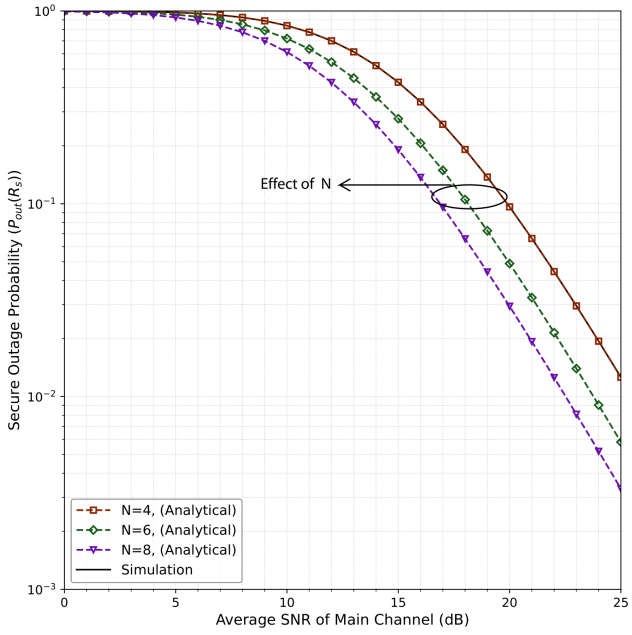


Fig. 6: The impact of N on SOP for $\kappa = 1$, $\mu = 1$ and $\bar{\gamma}_e = 1$dB.

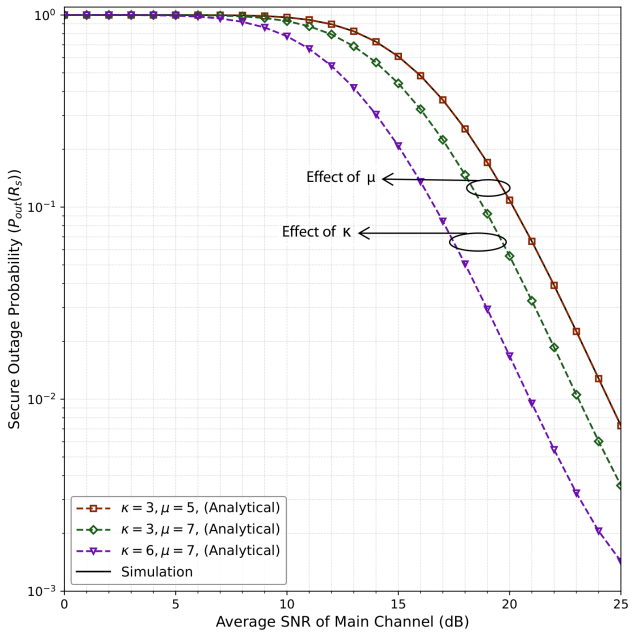


Fig. 7: Variation of SOP with respect to κ and μ when N=4, and $\bar{\gamma}_e$=1 dB.

eavesdropper's interference capability.

Fig. 6 shows the SOP versus the average SNR of the main channel for $N = 4, 6, 8$. The SOP is plotted over an extended SNR range of 0–25 dB to clearly illustrate the high-SNR behavior of the secure outage performance. The SOP decreases rapidly with increasing SNR, and a larger N consistently achieves lower SOP at any given SNR level. For instance, at an SNR of 20 dB, increasing N from 4 to 8 reduces the SOP by approximately one order of magnitude, demonstrating the strong impact of RIS element count on secrecy reliability.

Fig. 7 demonstrates the joint impact of κ and μ on SOP. Three parameter combinations are considered: $(\kappa = 3, \mu = 5)$, $(\kappa = 3, \mu = 7)$, and $(\kappa = 6, \mu = 7)$. It is seen that when μ increases from 5 to 7 with fixed $\kappa = 3$, there is a significant decrease in SOP due to the advantage of having more multipath diversity. Further increasing κ from 3 to 6 with $\mu = 7$ yields additional secrecy improvement, confirming

TABLE II: Quantitative Impact of N on Secrecy Metrics of the Proposed Scheme

Metric (at SNR)	$N = 4$	$N = 6$	$N = 8$
PNZSC ($\bar{\gamma}_d = 0$ dB)	0.078	0.166	0.254
PNZSC ($\bar{\gamma}_d = 5$ dB)	0.41	0.61	0.72
IP ($\bar{\gamma}_d = 5$ dB)	0.58	0.39	0.27
SOP ($\bar{\gamma}_d = 20$ dB)	9.6×10^{-2}	4.9×10^{-2}	2.95×10^{-2}

that both stronger LOS components and richer multipath conditions contribute positively to secrecy performance.

Table II quantifies the secrecy improvement achieved by increasing N in the proposed model. In [7], raising the number of reflecting elements from 25 to 35 yields only a 3 dB SNR gain at a fixed outage target, and from 35 to 60 a further 4 dB. The proposed fully RIS-dependent scheme reaches substantial secrecy improvement with a far smaller element count ($N = 4$ to 8), since the absent direct link no longer dilutes the RIS beamforming gain. The RIS-NOMA scheme in [9] faces a different limitation: increasing N lowers the outage probability but raises the intercept probability, a reliability-security trade-off inherent to that design. No such trade-off appears here. All three secrecy metrics, PNZSC, IP, and SOP improve together as N increases, because the eavesdropper's channel receives no benefit from the phase alignment intended for the destination. The relay-based multicast scheme in [14] shows the reverse behavior for the LOS parameters κ and μ: raising either one reduces the secrecy multicast capacity. In the proposed RIS-only setting, raising κ and μ instead strengthens PNZSC.

Evaluating (12), (14), and (16) requires truncating the infinite series to $K = 20$ terms, which gives an accuracy of 10^{-6}. The series terms decay rapidly with k_1 and k_2, so each SNR point can be computed efficiently on a standard desktop processor. This confirms the practical value of the derived expressions for offline system design and link-budget analysis. Real-time implementation would still need further optimization of the series truncation.

V. Conclusion

In this study, a novel analytical model incorporating an RIS-based approach is developed to enhance physical layer security, and its secrecy performance is evaluated. Specifically, series-form expressions for three secrecy metrics namely, PNZSC, IP, and SOP have been established based on the statistical behavior of the considered fading model. Monte Carlo simulations are employed to validate the analytical results. It is observed that the higher RIS reflecting elements, N, significantly improves all three secrecy metrics by strengthening the legitimate channel and reducing signal leakage to the eavesdropper. Likewise, higher values of the Rician factor, κ, and the multipath clustering parameter, μ, also enhanced secrecy performance. This confirmed the benefits of stronger line-of-sight (LOS) components and richer multipath diversity in securing the communication link. The strong consistency between analytical and simulation results across all figures confirms the validity of the derived results. These findings highlight RIS as an effective and energy-efficient method for achieving physical layer security in various fading environments without raising transmit power. Future research may explore imperfect channel state information, correlated fading, hardware limitations, and multiple eavesdropper scenarios to further develop this analysis.

References

[1] L. Yang, J. Yang, W. Xie, M. O. Hasna, T. Tsiftsis, and M. D. Renzo, "Secrecy performance analysis of RIS-aided wireless communication systems," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 10, pp. 12 296–12 300, 2020.

[2] A.-A. A. Boulogeorgos and A. Alexiou, "Performance analysis of reconfigurable intelligent surface-assisted wireless systems and comparison with relaying," *IEEE Access*, vol. 8, pp. 94 463–94 483, 2020.

[3] Q. Wu, S. Zhang, B. Zheng, C. You, and R. Zhang, "Intelligent reflecting surface-aided wireless communications: A tutorial," *IEEE Transactions on Communications*, vol. 69, no. 5, pp. 3313–3351, 2021.

[4] D.-T. Do, A.-T. Le, and S. Mumtaz, "Secure performance analysis of RIS-aided wireless communication systems," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2021, pp. 1–6.

[5] A. K. Yadav, S. Yadav, M. K. Shukla, D. S. Gurjar, and X. Li, "Secrecy performance analysis of RIS-enabled wireless networks over Rayleigh fading channels," in *Proc. Advanced Communication Technologies and Signal Processing (ACTS)*, 2021, pp. 1–6.

[6] J. F. Paris, "Statistical characterization of κ-μ shadowed fading," *IEEE Transactions on Vehicular Technology*, vol. 63, no. 2, pp. 518–526, 2014.

[7] A. Al-Rimawi and A. Al-Dweik, "On the performance of RIS-assisted communications with direct link over κ-μ shadowed fading," *IEEE Open Journal of the Communications Society*, vol. 3, pp. 2314–2328, 2022.

[8] H. V. K. Huu Q. Tran, "Performance analysis for reconfigurable intelligent surface-aided communication systems with energy harvesting under imperfect nakagami-m channel information," *EAI Endorsed Transactions on Industrial Networks and Intelligent Systems*, vol. 11, no. 1, 2024.

[9] M. Li, K. Ren, F. E. Bouanani, W. Chen, and Z. Han, "Secure performance of RIS-NOMA system under κ-μ shadowed fading channel," *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3543–3547, 2024.

[10] A. Sikri, A. Mathur, and G. Kaddoum, "Joint impact of phase error, transceiver hardware impairments, and mobile interferers on RIS-aided wireless system over κ-μ fading channels," *IEEE Communications Letters*, vol. 26, no. 10, pp. 2312–2316, 2022.

[11] M. A. A. Mamun, M. Z. I. Sarkar, A. Alam, and M. R. A. Ruku, "Performance analysis of reconfigurable intelligent surface in enhancing security of wireless networks," in *Proc. 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 2024, pp. 1–5.

[12] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. New York, NY, USA: Dover Publications, 1964.

[13] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 7th ed. San Diego, CA: Academic Press, 2007.

[14] A. Alam and M. Z. I. Sarkar, "Enhancement of multicast security with opportunistic relaying technique over κ-μ shadowed fading channels," *Journal of Computer and Communications*, vol. 10, pp. 121–137, November 2022.